Header

[redacted]
Lab Number: 3
Lab Name: Digital Design using FPGAs

Task

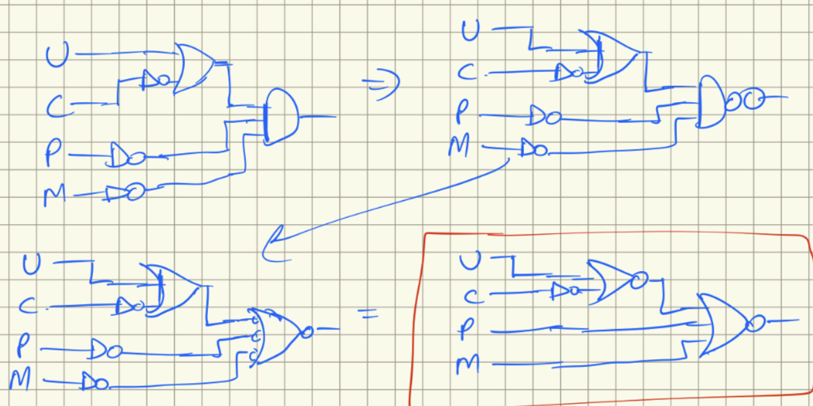
Truth table

$M=1$ (Default), $\overline{M}=0$

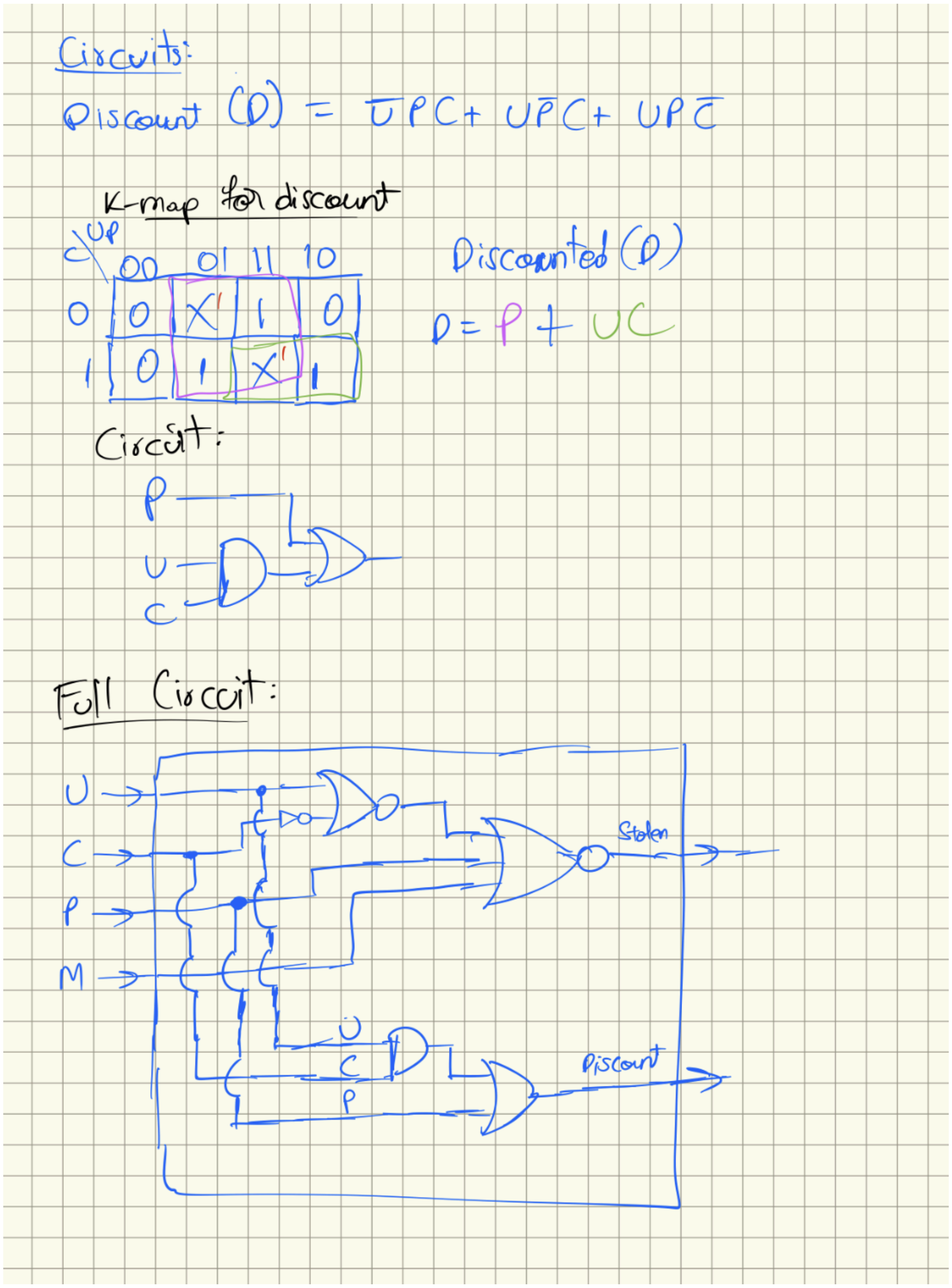
U	C	P	Expensive	Stolen (M=0)	Stolen (M=1)
0	0	0	Yes	1	0
0	0	1	No	0	X
0	1	1	No	0	X
1	0	0	Yes	1	0
1	0	1	Yes	1	0
1	1	0	No	0	X

CM \ UP	00	01	11	10
00	1	X	0	1
01	0	X	X	0
11	X	X	X	0
10	0	0	X	1

$$S = U\overline{P}\,\overline{M} + \overline{P}\,\overline{C}\,\overline{M}$$
$$= \overline{M}\,\overline{P}(U+\overline{C})$$

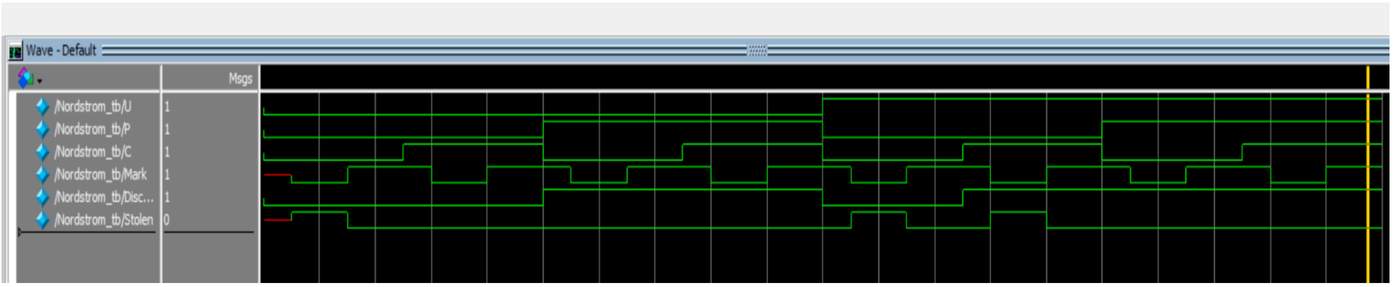


Above screenshot shows the truth table, karnaugh map, and logic simplification work I did to arrive at the final minimized stolen detector circuit.



This screenshot shows the k-map and simplified discount circuit as well as the integrated Nordstrom module - the full circuit which has the stolen and discount circuits

Modelsim Wave view:



The screenshot shows the simulated graph from the testbench. The testbench tests all the inputs possible - 3 bits for UPC and 2 bits (1 and 0) for Mark for each UPC code combination.

All possible inputs are modelled in this wave diagram.

When expensive item UTC codes are inputted, the stolen signal is either 1 or 0 which is dependent on the Mark signal being 1 or 0 - Mark being 1 on an expensive item indicates that it is not stolen, and the waveform shows the stolen is 0 in that scenario.
Similarly, whenever an item with a discounted UTC code is inputted, the discounted signal is actuated regardless of the stolen signal.

The above waveform captures the behavior of the module. The waveform demonstrates simulated behavior to perfectly align with the specification for this module as described in the lab, especially for the expected inputs.

Time spent: 5 hours